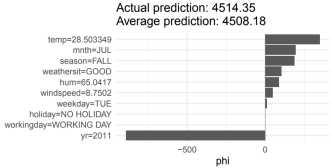
Interpretable Machine Learning

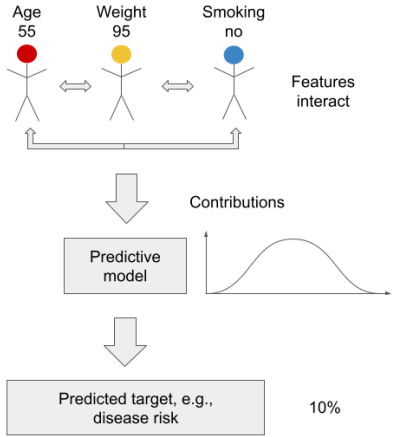
Shapley Values for Local Explanations

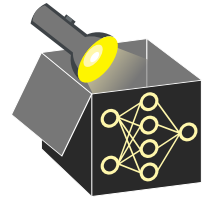


Learning goals

- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

FROM GAME THEORY TO MACHINE LEARNING





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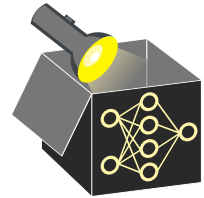

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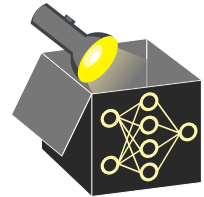

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- Players: Features $x_j, j \in \{1, \ldots, p\}$ which cooperate to produce a prediction
  $\rightsquigarrow$ How can we predict with a subset of features without changing the model?
  $\rightsquigarrow$ PD function ("removing" by marginalizing): $\hat{f}_S(\mathbf{x}_S) := \int_{X_{-S}} \hat{f}(\mathbf{x}_S, X_{-S}) d\mathbb{P}_{X_{-S}}$

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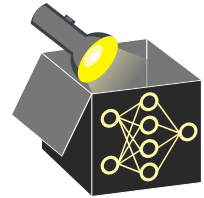

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- For $S = \emptyset$ $\quad \hat{f}_S(\mathbf{x}_S) = \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ (average over all possible values of all the features) $\rightsquigarrow$ Can't be used as a value function since $v(\emptyset)$ must be 0 but $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ can be non zero.
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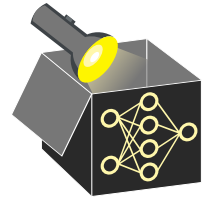

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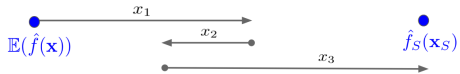

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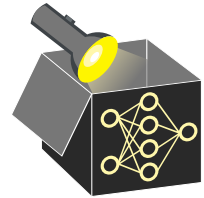

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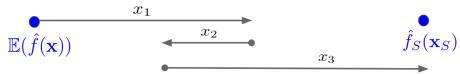
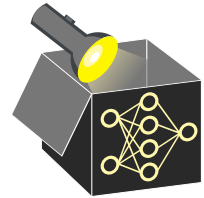


- Marginal contribution: $v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$
  $\rightsquigarrow$ $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ cancels out due to the subtraction of value functions

SHAPLEY VALUE - DEFINITION

Shapley (1953) | Strumbelj et al. (2014)



Shapley value ϕ_j of feature j for observation $\mathbf{x}$ via **order definition**:

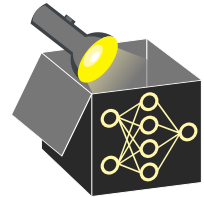
$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\mathbf{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})}_{\text{marginal contribution of feature } j}$$

- Interpretation: Feature x_j contributed ϕ_j to difference between $\hat{f}(\mathbf{x})$ and average prediction
  $\rightsquigarrow$ Note: Marginal contributions and Shapley values can be negative
- For exact computation of $\phi_j(\mathbf{x})$, the PD function $\hat{f}_S(\mathbf{x}_S) = \frac{1}{n}\sum_{i=1}^n \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}^{(i)})$ for any set of features S can be used which yields

$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \frac{1}{n} \sum_{i=1}^{n} \hat{f}(\mathbf{x}_{S_j^\tau \cup \{j\}}, \mathbf{x}^{(i)}_{-\{S_j^\tau \cup \{j\}\}}) - \hat{f}(\mathbf{x}_{S_j^\tau}, \mathbf{x}^{(i)}_{-S_j^\tau})$$

$\rightsquigarrow$ Note: $\hat{f}_S$ marginalizes over all other features $-S$ using all observations $i = 1, \ldots, n$

ESTIMATION: A PRACTICAL PROBLEM



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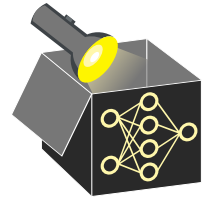

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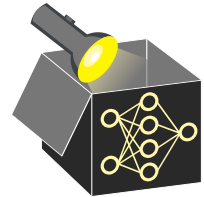

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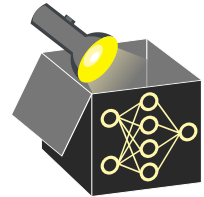

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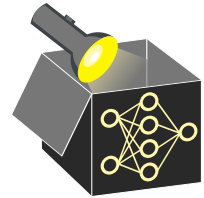
- M is a tradeoff between accuracy of the Shapley value and computational costs
  $\rightsquigarrow$ The higher M, the closer to the exact Shapley values, but the more costly the computation

APPROXIMATION ALGORITHM

Strumbelj et al. (2014)



Estimation of ϕ_j for observation $\mathbf{x}$ of model $\hat{f}$ fitted on data $\mathcal{D}$ using sample size M:

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APPROXIMATION ALGORITHM

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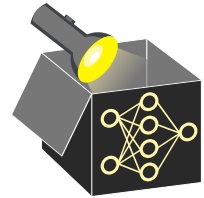

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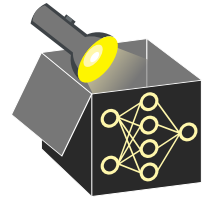

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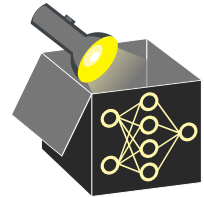

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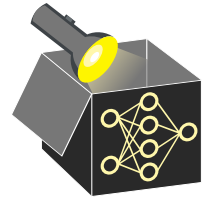

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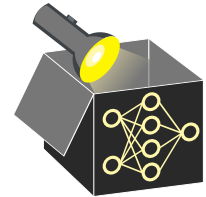

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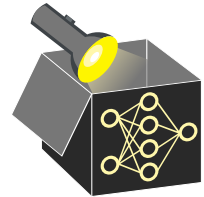

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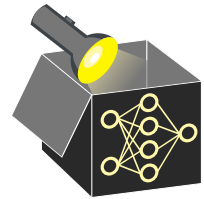

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   5. Compute difference $\phi_j^m = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
      $\rightsquigarrow \hat{f}_{S_m}(\mathbf{x}_{S_m})$ is approximated by $\hat{f}(\mathbf{x}_{-j}^{(m)})$ and $\hat{f}_{S_m \cup \{j\}}(\mathbf{x}_{S_m \cup \{j\}})$ by $\hat{f}(\mathbf{x}_{+j}^{(m)})$ over M iters.

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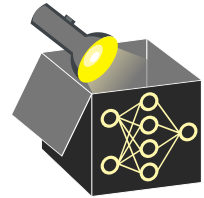

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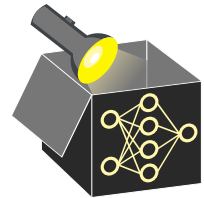
      - $\mathbf{x}_{+j}^{(m)} = (\underbrace{x_{\tau(1)}, \ldots, x_{\tau(|S_m|)}, x_j}_{\mathbf{x}_{S_m \cup \{j\}}}, \underbrace{z_{\tau(|S_m|+2)}^{(m)}, \ldots, z_{\tau(p)}^{(m)}}_{\mathbf{z}_{-\{S_m \cup \{j\}\}}^{(m)}})$ takes features $S_m \cup \{j\}$ from $\mathbf{x}$
      - $\mathbf{x}_{-j}^{(m)} = (\underbrace{x_{\tau(1)}, \ldots, x_{\tau(|S_m|)}}_{\mathbf{x}_{S_m}}, \underbrace{z_j^{(m)}, z_{\tau(|S_m|+2)}^{(m)}, \ldots, z_{\tau(p)}^{(m)}}_{\mathbf{z}_{-S_m}^{(m)}})$ takes features S_m from $\mathbf{x}$ (takes $\{j\}$ randomly)
   5. Compute difference $\phi_j^m = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
      $\rightsquigarrow \hat{f}_{S_m}(\mathbf{x}_{S_m})$ is approximated by $\hat{f}(\mathbf{x}_{-j}^{(m)})$ and $\hat{f}_{S_m \cup \{j\}}(\mathbf{x}_{S_m \cup \{j\}})$ by $\hat{f}(\mathbf{x}_{+j}^{(m)})$ over M iters.
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \phi_j^m$

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



Definition

$\mathbf{x}$: obs. of interest

$\mathbf{x}$ with feature values in S_m (other are replaced)

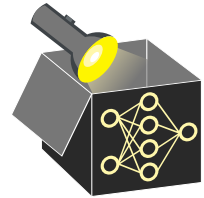
$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

$\mathbf{x}$ with feature values in $S_m \cup \{j\}$

	Temperature	Humidity	Windspeed	Year
x	10.66	56	11	2012
x_{+j}	10.66	56	$random: z_{windspeed}^{(m)}$	2012
x_{-j}	10.66	56	$random: z_{windspeed}^{(m)}$	$random: z_{year}^{(m)}$

j

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



Definition

Contribution of feature j to coalition S_m

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\underbrace{\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]}_{:=\Delta(j,S_m)}$$

- $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$ is the marginal contribution of feature j to coalition S_m
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{temp, hum\}$

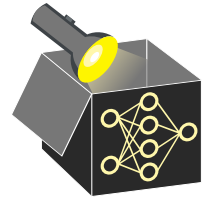
	Temperature	Humidity	Windspeed	Year	Count
x	10.66	56	11	2012	5600
x_{+j}	10.66	56	$random: z_{windspeed}^{(m)}$	2012	4900
x_{-j}	10.66	56	$random: z_{windspeed}^{(m)}$	$random: z_{year}^{(m)}$	

j $\hat{f}$

700

$\Delta(j, S_m)$ marginal contribution

SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

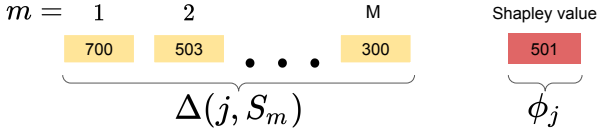


Definition

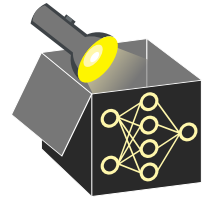
average the contributions of feature j

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

- Compute marginal contribution of feature j towards the prediction across all randomly drawn feature coalitions $S_1, \ldots, S_m$
- Average all M marginal contributions of feature j
- Shapley value ϕ_j is the payout of feature j, i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific observation $\mathbf{x}$



REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS



We take the general axioms for Shapley Values and apply it to predictions:

- **Efficiency**: Shapley values add up to the (centered) prediction: $\sum_{j=1}^{p} \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}(\hat{f}(X))$

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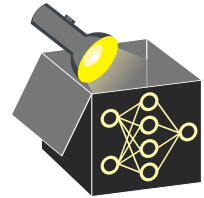

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- **Symmetry**: Two features j and k that contribute the same to the prediction get the same payout
  $\rightsquigarrow$ interaction effects between features are fairly divided
  $\hat{f}_{S\cup\{j\}}(\mathbf{x}_{S\cup\{j\}}) = \hat{f}_{S\cup\{k\}}(\mathbf{x}_{S\cup\{k\}})$ for all $S \subseteq P \setminus \{j, k\}$ then $\phi_j = \phi_k$

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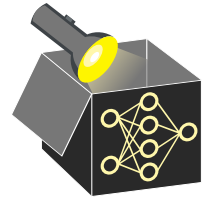

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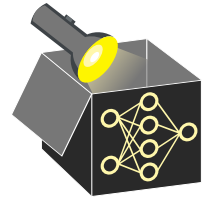

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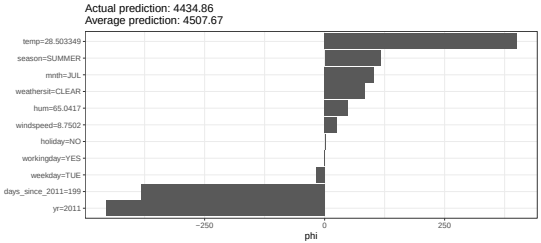
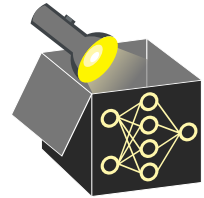
- **Additivity**: For a prediction with combined payouts, the payout is the sum of payouts: $\phi_j(v_1) + \phi_j(v_2) \rightsquigarrow$ Shapley values for model ensembles can be combined

BIKE SHARING DATASET



- Shapley values of observation $i = 200$ from the bike sharing data
- Difference between model prediction of this observation and the average prediction of the data is fairly distributed among the features (i.e., $4434 - 4507 \approx -73$)
- Feature value temp = 28.5 has the most positive effect, with a contribution (increase of prediction) of about +400

ADVANTAGES AND DISADVANTAGES



Advantages:

- **Solid theoretical foundation** in game theory
- Prediction is **fairly distributed** among the feature values $\rightsquigarrow$ easy to interpret for a user
- **Contrastive explanations** that compare the prediction with the average prediction

Disadvantages:

- Without sampling, Shapley values need a lot of computing time to inspect all possible coalitions
- Like many other IML methods, Shapley values suffer from the inclusion of unrealistic data observations when features are correlated